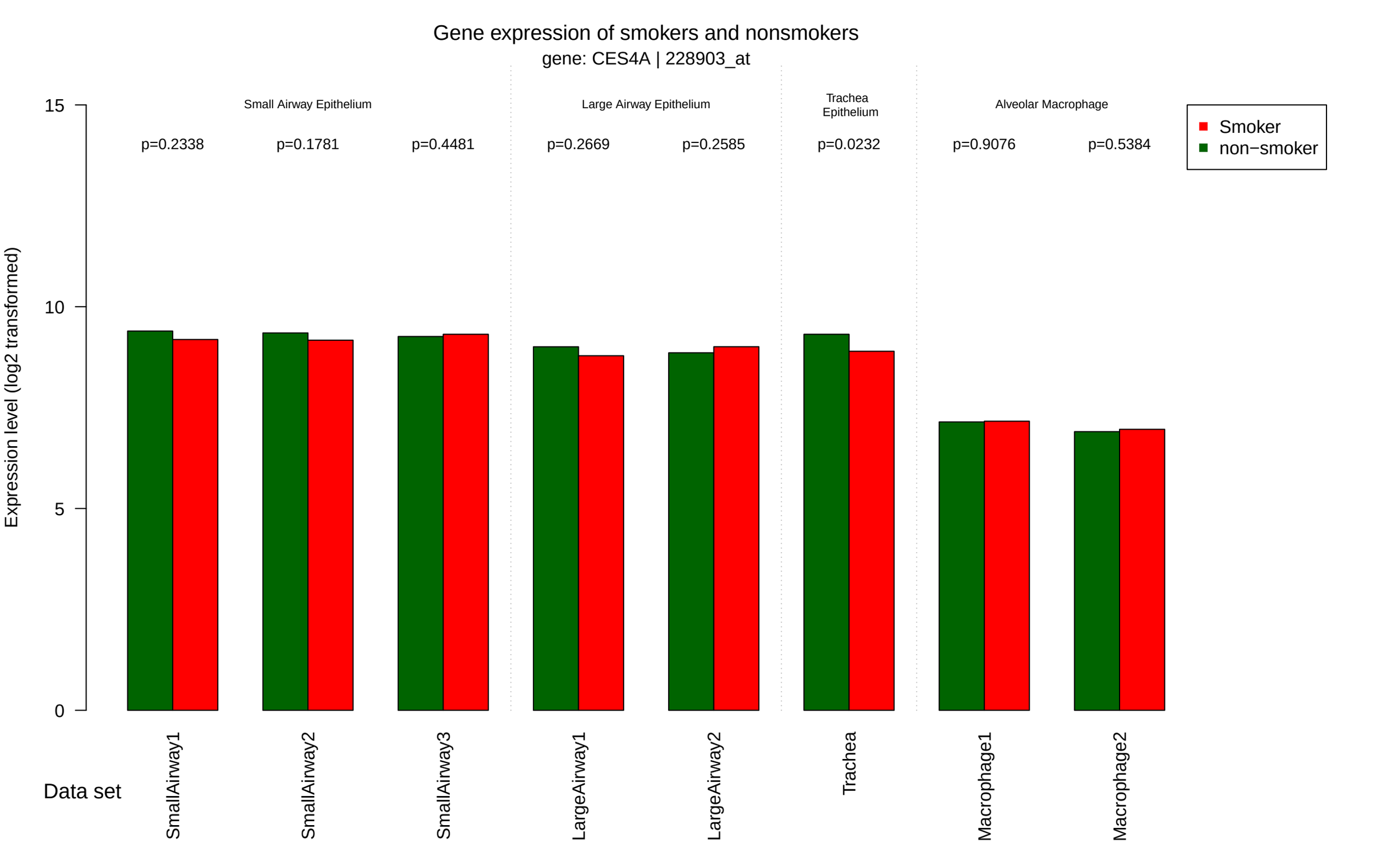
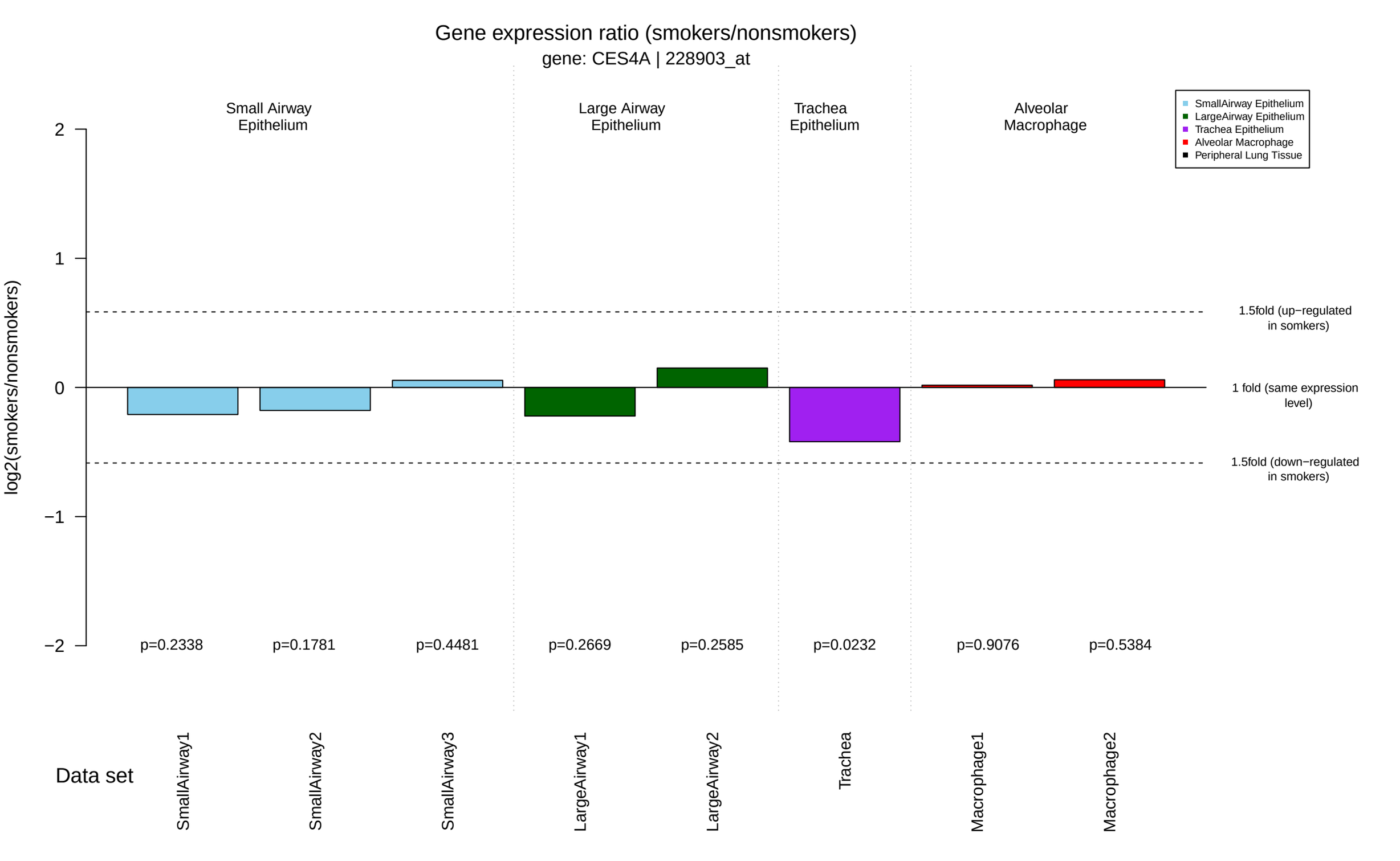
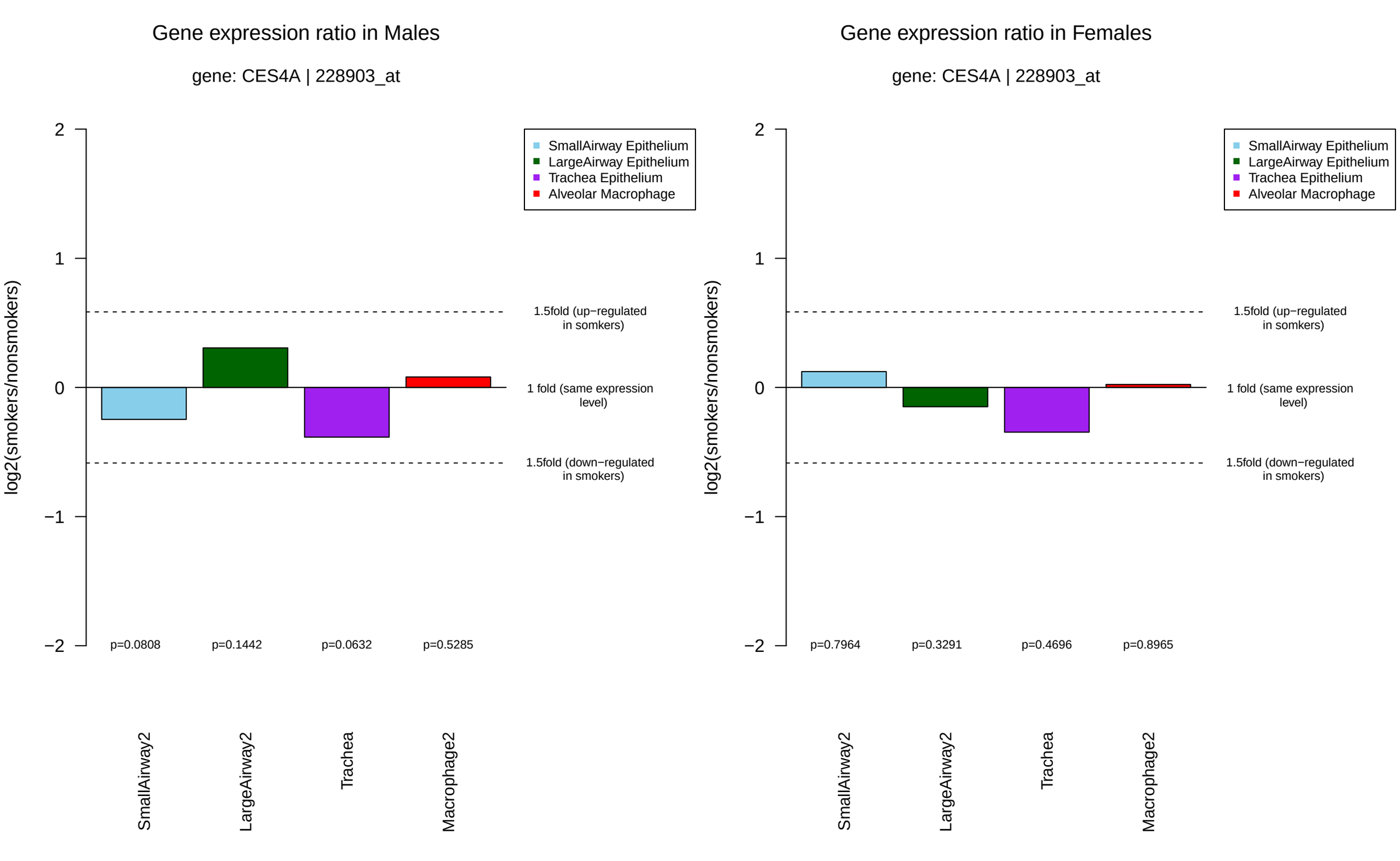
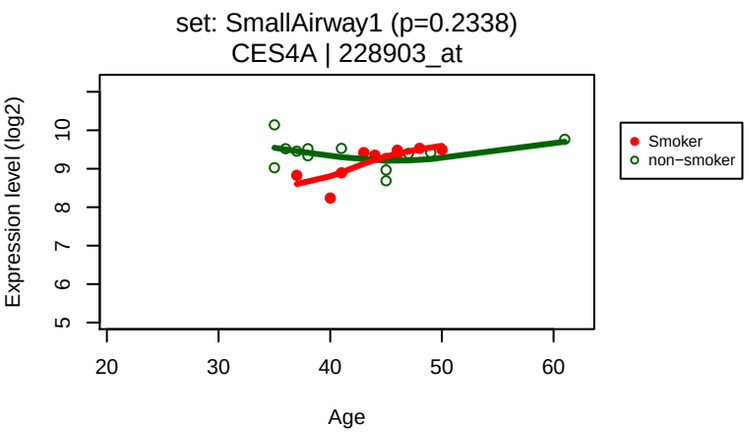
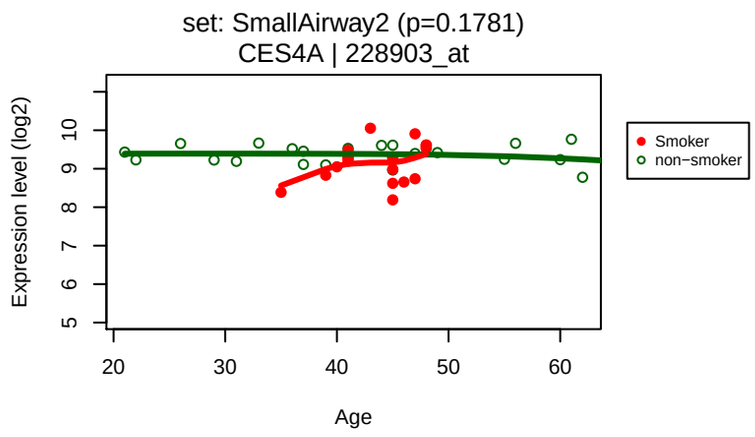
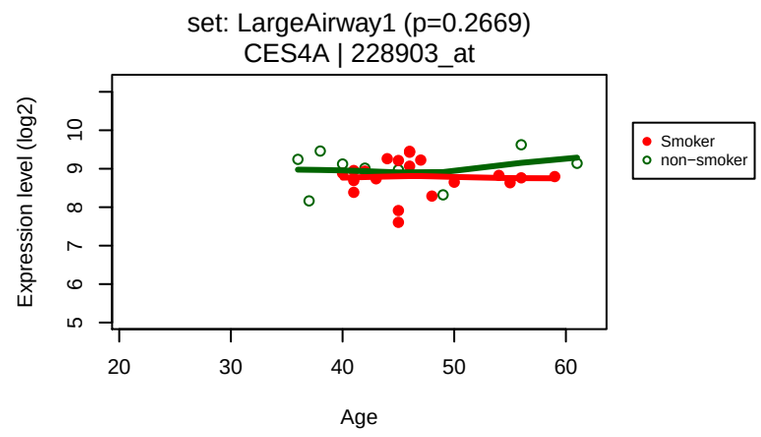

Gene expression of smokers and nonsmokers
gene: CES4A | 228903_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2338
p=0.1781
p=0.4481
p=0.2669
p=0.2585
p=0.0232
p=0.9076
p=0.5384
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

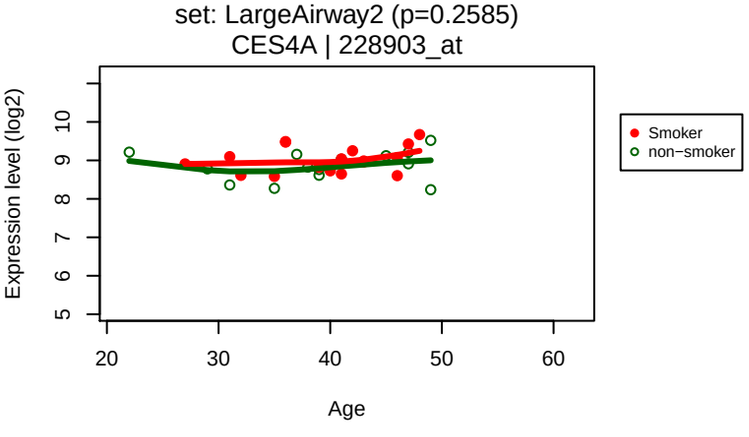

Gene expression ratio (smokers/nonsmokers)
gene: CES4A | 228903_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2338
p=0.1781
p=0.4481
p=0.2669
p=0.2585
p=0.0232
p=0.9076
p=0.5384
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

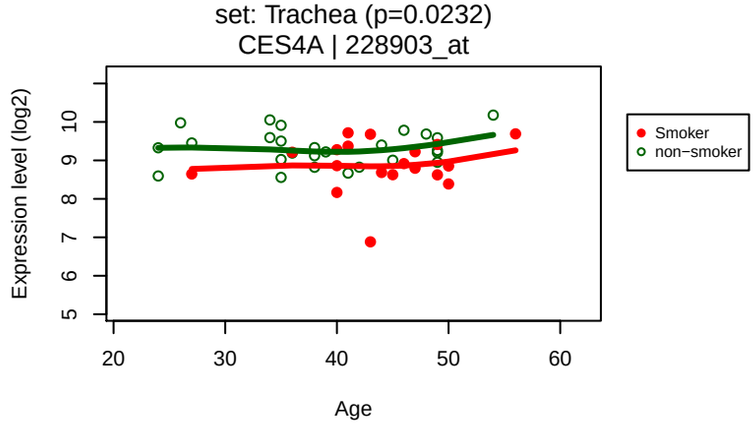

Gene expression ratio in Males
gene: CES4A | 228903_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0808
p=0.1442
p=0.0632
p=0.5285
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CES4A | 228903_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7964
p=0.3291
p=0.4696
p=0.8965
SmallAirway2
LargeAirway2
Trachea
Macrophage2

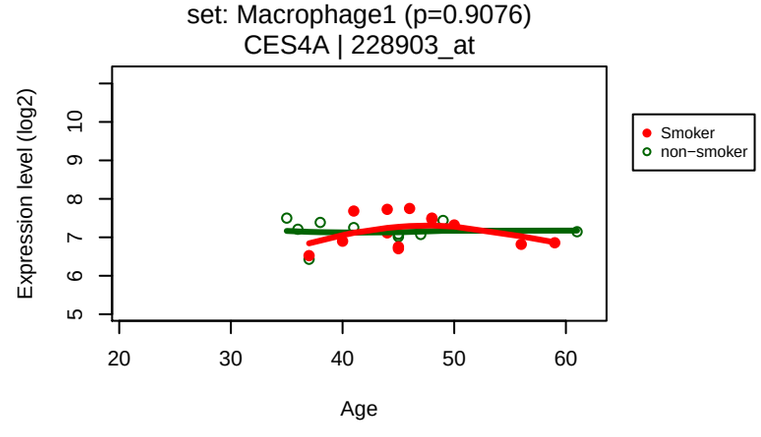

set: SmallAirway1 (p=0.2338)
CES4A | 228903_at
Expression level (log2)
Age
Smoker
non−smoker

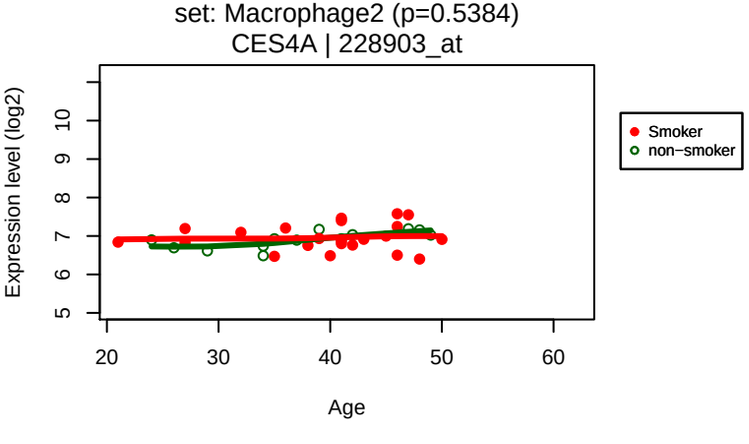

set: SmallAirway2 (p=0.1781)
CES4A | 228903_at
Expression level (log2)
Age
Smoker
non−smoker

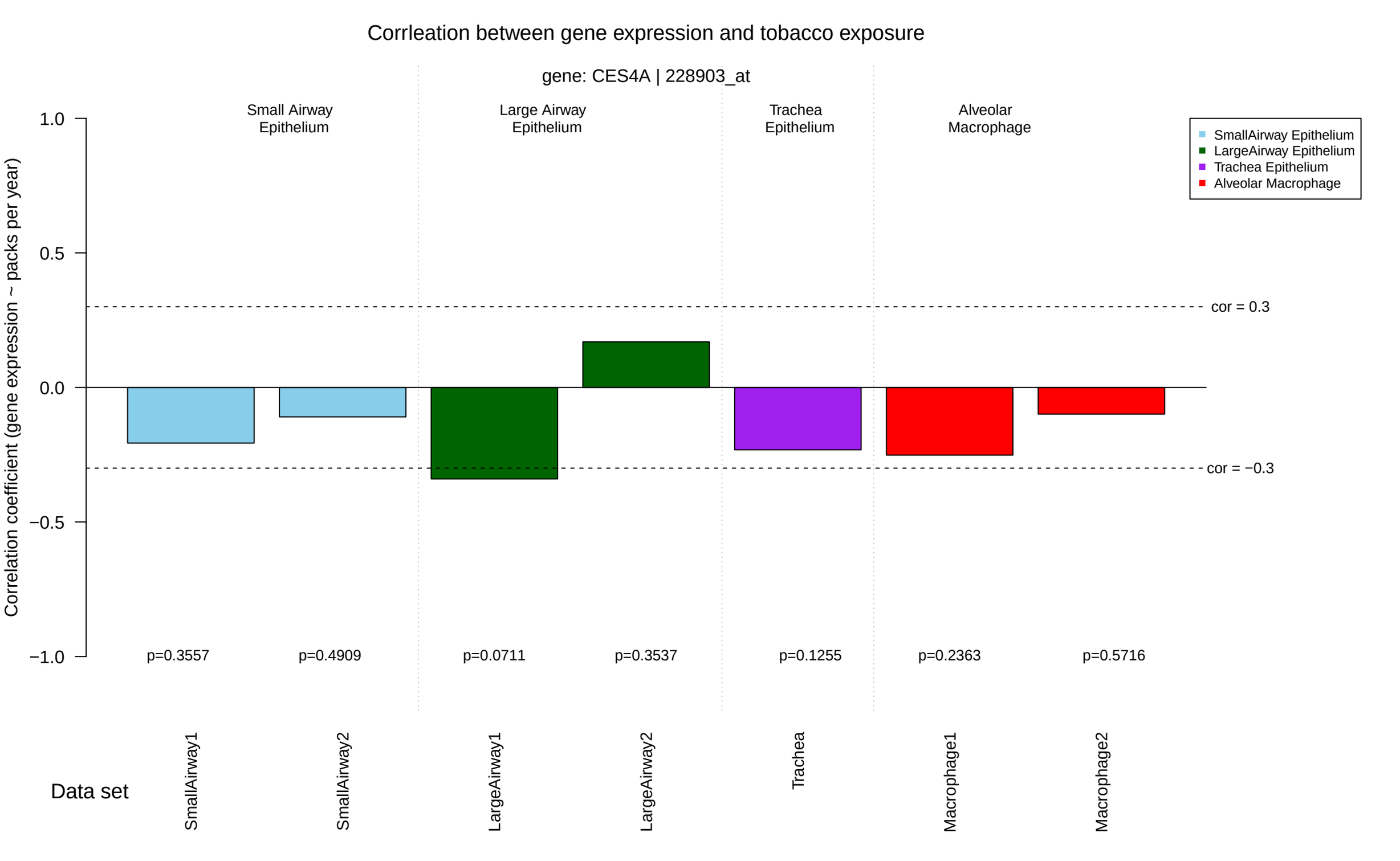

set: LargeAirway1 (p=0.2669)
CES4A | 228903_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2585)
CES4A | 228903_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0232)
CES4A | 228903_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9076)
CES4A | 228903_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5384)
CES4A | 228903_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CES4A | 228903_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3557
p=0.4909
p=0.0711
p=0.3537
p=0.1255
p=0.2363
p=0.5716
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2